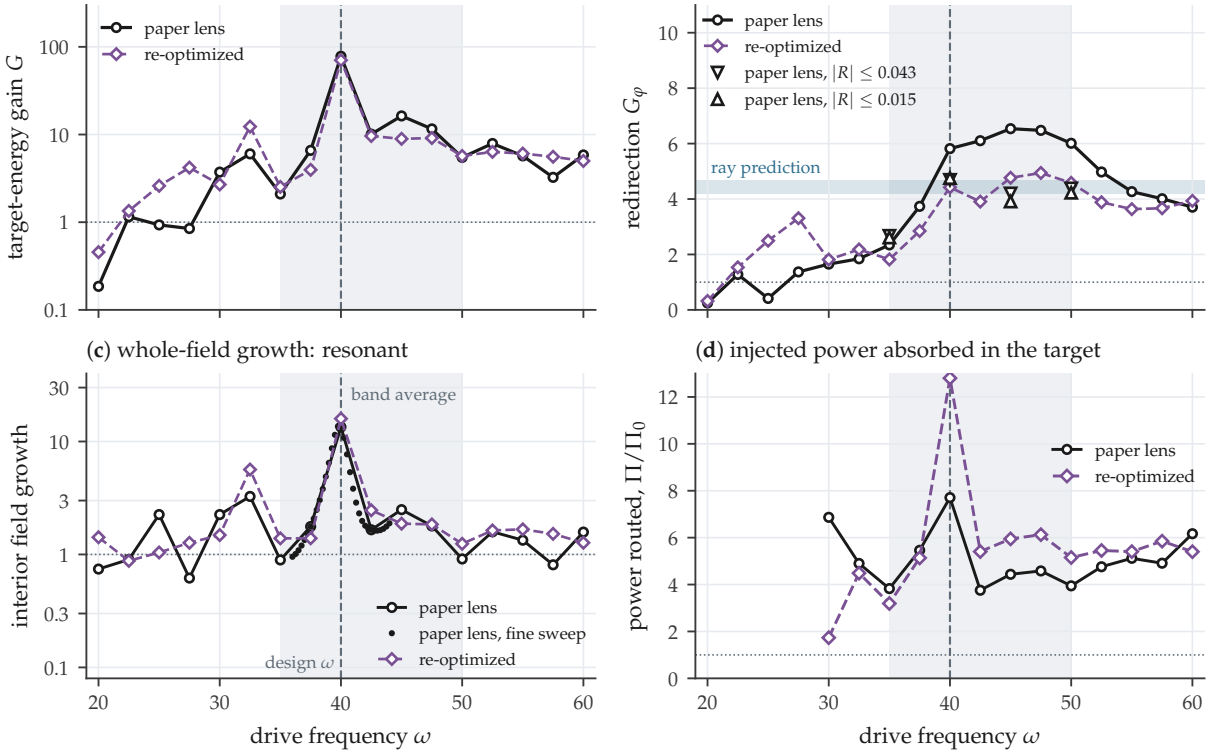

(a) single-frequency gain: narrow peak
paper lens
re-optimized
target-energy gain $G$
100
10
1
0.1
(b) redirection: broadband
paper lens
re-optimized
paper lens, $|R| \leq 0.043$
paper lens, $|R| \leq 0.015$
ray prediction
redirection $G_\varphi$
10
8
6
4
2
0
(c) whole-field growth: resonant
band average
paper lens
paper lens, fine sweep
re-optimized
design $\omega$
interior field growth
30
10
3
1
0.3
0.1
20
30
40
50
60
drive frequency $\omega$
(d) injected power absorbed in the target
paper lens
re-optimized
power routed, $\Pi/\Pi_0$
12
10
8
6
4
2
0
20
30
40
50
60
drive frequency $\omega$